

Project name: Current Status of Electric Vehicles in the US market

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Project description: When we plan to purchase a new car, there are more choices nowadays compared to a few years ago. As electric vehicles are introduced to the market, more people are choosing electric vehicles over traditional gasoline cars due to environmental or price reasons. In this project, I aim to analyze the EV adoption factors and EV adoption rates within the current US local market. Specifically, what factor is impacting EV adoption rate in US market.

Source 1: American Community Survey 1-Year Data (2023) (api)

Link: <https://www.census.gov/data/developers/data-sets/acs-1year.2023.html>

Cite: U.S. Census Bureau. (2023). *American Community Survey 1-year data (2023)*.

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Data will be collected using the U.S. Census API by making HTTP requests with selected variables, such as income and total population. The API returns data in JSON format, which will be converted into a Pandas DataFrame for further processing and merged at the state level.

Source 2: Alternative Fueling Historical station counts (EV charging stations) (web extraction)

Link: <https://afdc.energy.gov/files/docs/historical-station-counts.xlsx?year=2023>

Cite: U.S. Department of Energy, Alternative Fuels Data Center. (2023). Historical station counts (EV charging stations).

<https://afdc.energy.gov/files/docs/historical-station-counts.xlsx?year=2023>

This dataset provides EV charging infrastructure data by state, including the number of charging stations and charging ports. The data can be extracted programmatically using `pandas.read_html()` from the webpage tables.

Source 3: Electric vehicle registrations by state (xlsx)

Link: <https://afdc.energy.gov/vehicle-registration?year=2023>

Cite: U.S. Department of Energy, Alternative Fuels Data Center. (2023). Electric vehicle registrations by state. <https://afdc.energy.gov/vehicle-registration?year=2023>

This dataset can be downloaded from the web in an `.xlsx` format, which could be processed by the `pandas` library. By using `pandas`, I can analyze the car under fuel type, which counts to the state level.

Data combination: By using the `state` as the primary key, all datasets will be merged into a unified dataframe structure. The vehicle registrations dataset will be aggregated to the state level to show EV and gasoline vehicle counts. The ACS dataset will provide corresponding state-level population and income data, while the Alternative Fueling Station Locator dataset will provide information on EV charging stations. After standardizing state names across all datasets, they will be combined into a single dataset for analysis. This combined dataset will allow the project to examine EV adoption trends across the U.S. and evaluate whether electric vehicles are replacing gasoline vehicles at the state level.